Geotechnical Investigation

1. Drilling & Sampling

- Drilling and sampling
  - Drilling and sampling is a geotechnical investigation method used to extract soil or rock samples from the subsurface to determine their engineering, geological, and environmental properties. This process involves boring holes (drilling) and collecting materials (sampling) for laboratory testing, crucial for evaluating site strength, contamination, and design parameters
- Soil Boring
  - Soil boring is a geotechnical drilling process that extracts subsurface soil and rock samples to analyse soil layers, strength, and composition. It is essential for determining foundation requirements, assessing groundwater levels, and evaluating site suitability for construction. These samples are analysed for structural design and environmental assessment
- Core Drilling
  - Core drilling is a specialized rotary drilling technique that uses a hollow, diamond-tipped or carbide bit to cut a perfectly cylindrical hole, extracting a solid "core" sample of material rather than grinding it into dust
- Standard Penetration Test
  - The Standard Penetration Test (SPT) is a common in-situ geotechnical field test used to determine the density, strength, and bearing capacity of soil by driving a split-spoon sampler into the ground.

- Auger Boring
  - The auger boring method of soil exploration involves rotating a helical screw (auger) into the ground to create shallow to moderate-depth boreholes, primarily in soft to stiff cohesive soils above the water table. It is a fast, economical, and popular method for collecting disturbed soil samples. Hand-operated augers reach depths up to 6m,
- Collecting Distributed and Undistributed Samples
  - Collecting disturbed and undisturbed soil samples is a fundamental aspect of geotechnical engineering investigations, designed to assess the engineering properties of soil layers at different depths for construction design. Disturbed samples are used to classify soil and identify its grain properties, while undisturbed samples are critical for determining the in-situ structure, strength, and settlement behavior of the soil.
- SBC – Safe Bearing Capacity
  - Safe Bearing Capacity (SBC) is the maximum pressure a soil can safely withstand without experiencing shear failure or excessive settlement. It is a critical geotechnical parameter, calculated by dividing the ultimate bearing capacity by a factor of safety
- Dynamic Cone Penetration Test
  - The Dynamic Cone Penetration Test (DCPT) is a rapid, in-situ geotechnical test used to evaluate the strength and consistency of soil layers and compacted materials.
- Cone Penetration Test
  - The Cone Penetration Test (CPT) is a widely used, rapid, and accurate in-situ geotechnical method for determining subsurface soil conditions and properties.

- Trial Pits
  - A trial pit (or test pit) is a shallow, excavated hole—typically 1–5 meters deep—used to directly observe, sample, and log subsurface soil, rock, and groundwater conditions before construction.

2. Foundation Recommendations

- Shallow Foundations
- Ground Improvement Technique
- Deep Foundations - Pile Foundations

3. Specialised Field Testing

- Electric Resistivity Test / Soil Resistivity Test
  - An electrical resistivity test of soil measures how strongly soil resists the flow of electric current, It is a critical, non-destructive geotechnical method used to evaluate soil composition, moisture content, and compaction, mainly for designing electrical grounding systems.
- Vane Shear Test
  - The vane shear test is a geotechnical method used to determine the in-situ or laboratory undrained shear strength of soft to medium-stiff cohesive soils (clays and silts). It is especially useful for determining the shear strength of sensitive clays, as it measures both the undisturbed and remoulded strength to calculate sensitivity
- Field permeability test
  - Field permeability tests determine the in-situ hydraulic conductivity of soil (how easily water flows through pores), crucial for foundation design and hydrogeological studies
- Field Sand replacement test
  - The Field Sand Replacement test is a geotechnical method to determine the *in-situ* dry density of compacted soil.

- Field core cutter test
  - The Field Core Cutter Test is a standard geotechnical field method used to determine the *in-situ* dry density and compaction level of soft, cohesive soils (e.g., clays, silts). By driving a calibrated steel cylinder into the ground, it measures the density in place, ensuring soil layers meet construction quality standards
- Chemical analysis of water (Ground/ Bore) For construction purpose
  - measuring physical and chemical parameters (NAOH, H2SO4, Cl, SO4, Inorganic Solids, Suspended matter, Organic Solids, pH Value) etc,
- Chemical analysis of water (Ground/ Bore) For drinking purpose
  - measuring physical and chemical parameters (pH, Colour, Odour, Taste, Turbidity, Total Dissolved Solids, Ba, Cu, Fe, Mn, NO3, F, Zn, Ag, Al, Cl, Se, SO4, CaCO3, Ca, Mg, Na, Free Residual chlorine, Mineral oil, Boron as B, Arsenic as As, Lead as Pb, Cr, Ni) etc,

- Chemical analysis of Soil
  - measuring physical and chemical parameters (pH @ 1.0 % Suspension, Cl, SO4, N, PO4, Na) etc,

- Plate load test
  - The plate load test is a field method to determine a soil's ultimate bearing capacity and settlement behaviour under a given load. The Modulus of Subgrade Reaction – K value can be determined by conducting a plate load test

4. Laboratory Tests

- Specific gravity
  - The specific gravity It is a dimensionless, crucial engineering property used to determine void ratio, porosity, and degree of saturation.
- Sieve analysis
  - Dry Sieve Analysis
  - Dry Sieve Analysis is a laboratory method used to determine the grain size distribution of dry, free-flowing materials (such as sand, gravel, or powders) by passing them through a stacked series of sieves with progressively smaller mesh sizes.
  - Wet Sieve Analysis
  - Wet Sieve Analysis is a laboratory technique used to accurately determine the particle size distribution of materials—particularly soils, aggregates, or powders containing significant fines (clay/silt) or that are sticky.
- Atterberg's limits
  - Bulk density of soil, the mass of dry soil per unit volume (g/cm3) measures compaction and soil health. It is used to assess root growth restriction, calculate soil porosity, evaluate water infiltration capacity, and manage soil organic matter.
- Atterberg's limits
  - Atterberg limits are critical water content percentages (liquid, plastic, and shrinkage) that define the boundaries between solid, semi-solid, plastic, and liquid states of fine-grained soils (clays and silts).
- Natural moisture content.
  - Natural moisture content (or natural water content) is the ratio of the weight of water to the weight of solid particles in a soil sample, expressed as a percentage. It represents the actual amount of water present in the soil in its natural, field state, generally determined by oven-drying at 105^0C - 110^0C

- Direct shear test
  - The direct shear test is a laboratory method used to determine the shear strength parameters—cohesion (c) and angle of internal friction (φ) of soil by forcing a specimen to fail along a horizontal plane. It is particularly effective for testing granular, non-cohesive soils and provides quick results for geotechnical design applications
- Triaxial Test
  - A triaxial test is a common geotechnical laboratory test used to determine the shear strength parameters (cohesion and internal friction angle) of soil essential for designing foundations, slopes, and tunnels.
- Consolidation Test
  - The consolidation test measures the rate and magnitude of soil settlement in saturated, cohesive soils under vertical pressure.
- Hydrometer Analysis
  - Hydrometer analysis is a geotechnical laboratory method used to determine the particle size distribution of fine-grained soils (silts and clays) passing the #200 sieve (0.075 mm).
- Proctor compaction test
  - The Proctor compaction test is a laboratory method used to determine the maximum dry density (MDD) and optimum moisture content (OMC) of soil.
- Free Swell Index test
  - The Free Swell Index (FSI) test, per IS: 2720 (Part 40), measures the increase in volume of soil (10g , < 425-micron sieve) upon wetting, identifying its swelling potential.
- California Bearing Ratio Test (soaked & un soaked)
  - The California Bearing Ratio (CBR) test is a penetration test used to evaluate the mechanical strength of subgrade soil and base course materials for road and pavement design.

5. Tests On Rock

- Modulus of elasticity
- Core recovery & RQD Calculation
- Point load test
- Density
- Specific Gravity
- Allowable Bearing Pressure
- Unconfined compression test